



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 01/02/2018*

PROBLEM - We observe N samples of the following discrete-time real random process:

$$X[n] = A + B \sin(2\pi F_0 n) + W[n], \quad n = 0, 1, \dots, N-1,$$

where A and B are deterministic unknown parameters and $F_0 \in (0, 1/2)$ is a-priori known, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_w(e^{j2\pi f}) = \sigma_w^2$, where σ_w^2 is a-priori known.

- (1) Derive the joint probability density function (pdf) of the received data vector $\mathbf{X} \triangleq [X[0] \ X[1] \ \dots \ X[N-1]]^T$.
- (2) Under the assumption that $N \gg 1$, derive the joint Maximum Likelihood (ML) estimator of the parameters A and B , the pdf of the estimation errors $\varepsilon_A \triangleq A - \hat{A}_{ML}$ and $\varepsilon_B \triangleq B - \hat{B}_{ML}$, and the Mean Square Error (MSE) of the estimators, $MSE_A \triangleq E\{\varepsilon_A^2\}$ and $MSE_B \triangleq E\{\varepsilon_B^2\}$. Establish whether the estimators are biased and consistent.
- (3) Derive the Cramèr-Rao lower bound (CRB) on the estimation of A and B , compare them with the MSE of the ML estimators and establish whether the estimators are efficient or not.
- (4) Which one of the two parameters can be estimated with better accuracy? Provide an intuitive explanation. Which is the minimum value of N that guarantees that the standard deviation of ε_B is lower than the 10% of the true value of B under the assumption that $A=B=2$ and $\sigma_w^2=10$.



Solution

(1) The data can be modeled according to the deterministic linear model in AWGN:

$$\mathbf{X} = A \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + B \begin{bmatrix} \sin(2\pi F_0 0) \\ \sin(2\pi F_0 1) \\ \vdots \\ \sin(2\pi F_0 (N-1)) \end{bmatrix} + \mathbf{W} = A\mathbf{1} + B\mathbf{s} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

$$\boldsymbol{\theta}_{2 \times 1} \triangleq \begin{bmatrix} A \\ B \end{bmatrix}, \quad \mathbf{H}_{N \times 2} \triangleq [\mathbf{1} \quad \mathbf{s}]$$

$$\mathbf{X} \in N(\mathbf{A}\mathbf{1} + B\mathbf{s}, \sigma_w^2 \mathbf{I})$$

so $\mathbf{X}$ is a Gaussian random vector.

(2) For the deterministic linear model in AWGN we easily get:

$$\hat{\boldsymbol{\theta}}_{ML} = \mathbf{H}^\# \mathbf{X} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{X}$$

$$\mathbf{H}^\# = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T = \left(\begin{bmatrix} \mathbf{1}^T \\ \mathbf{s}^T \end{bmatrix} [\mathbf{1} \quad \mathbf{s}] \right)^{-1} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{s}^T \end{bmatrix} = \begin{bmatrix} \mathbf{1}^T \mathbf{1} & \mathbf{1}^T \mathbf{s} \\ \mathbf{s}^T \mathbf{1} & \mathbf{s}^T \mathbf{s} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{s}^T \end{bmatrix} = \begin{bmatrix} N & 0 \\ 0 & \frac{N}{2} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{1}^T \\ \mathbf{s}^T \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \\ \frac{2}{N} \mathbf{s}^T \end{bmatrix}$$

$$\hat{\boldsymbol{\theta}}_{ML} = \mathbf{H}^\# \mathbf{X} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \\ \frac{2}{N} \mathbf{s}^T \end{bmatrix} \mathbf{X} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \mathbf{X} \\ \frac{2}{N} \mathbf{s}^T \mathbf{X} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \sum_{n=0}^{N-1} X[n] \\ \frac{2}{N} \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n) \end{bmatrix}$$

where we exploited the fact that if $N \gg 1$, we have $\mathbf{1}^T \mathbf{1} = \|\mathbf{1}\|^2 = N$, $\mathbf{s}^T \mathbf{s} = \|\mathbf{s}\|^2 \cong \frac{N}{2}$, $\mathbf{1}^T \mathbf{s} = \mathbf{s}^T \mathbf{1} \cong 0$.

$$\hat{A}_{ML} = \frac{1}{N} \sum_{n=0}^{N-1} X[n]$$

$$\hat{B}_{ML} = \frac{2}{N} \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n)$$



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The estimators are **linear** and so the estimates are **Gaussian** random variables, being $\mathbf{X}$ a Gaussian vector; consequently the estimation errors are Gaussian random variables. To derive their mean values and their variances we can proceed as follows:

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{A}_{ML} \\ \hat{B}_{ML} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T \mathbf{X} \\ \frac{2}{N} \mathbf{s}^T \mathbf{X} \end{bmatrix} = \begin{bmatrix} \frac{1}{N} \mathbf{1}^T (\mathbf{A} \mathbf{1} + \mathbf{B} \mathbf{s} + \mathbf{W}) \\ \frac{2}{N} \mathbf{s}^T (\mathbf{A} \mathbf{1} + \mathbf{B} \mathbf{s} + \mathbf{W}) \end{bmatrix} = \begin{bmatrix} A + \frac{1}{N} \mathbf{1}^T \mathbf{W} \\ B + \frac{2}{N} \mathbf{s}^T \mathbf{W} \end{bmatrix}$$

$$\boldsymbol{\varepsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} A - \hat{A}_{ML} \\ B - \hat{B}_{ML} \end{bmatrix} = \begin{bmatrix} \varepsilon_A \\ \varepsilon_B \end{bmatrix} = \begin{bmatrix} -\frac{1}{N} \mathbf{1}^T \mathbf{W} \\ -\frac{2}{N} \mathbf{s}^T \mathbf{W} \end{bmatrix}$$

$$E\{\boldsymbol{\varepsilon}\} = \begin{bmatrix} E\{\varepsilon_A\} \\ E\{\varepsilon_B\} \end{bmatrix} = \begin{bmatrix} E\left\{-\frac{1}{N} \mathbf{1}^T \mathbf{W}\right\} \\ E\left\{-\frac{2}{N} \mathbf{s}^T \mathbf{W}\right\} \end{bmatrix} = \begin{bmatrix} -\frac{1}{N} \mathbf{1}^T E\{\mathbf{W}\} \\ -\frac{2}{N} \mathbf{s}^T E\{\mathbf{W}\} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Hence, the estimators are **unbiased**.

$$\begin{aligned} MSE\{\hat{\boldsymbol{\theta}}_{ML}\} &= \begin{bmatrix} MSE\{\hat{A}_{ML}\} \\ MSE\{\hat{B}_{ML}\} \end{bmatrix} = \begin{bmatrix} \text{var}\{\hat{A}_{ML}\} \\ \text{var}\{\hat{B}_{ML}\} \end{bmatrix} = \begin{bmatrix} \left(\frac{1}{N}\right)^2 \text{var}\{\mathbf{1}^T \mathbf{X}\} \\ \left(\frac{2}{N}\right)^2 \text{var}\{\mathbf{s}^T \mathbf{X}\} \end{bmatrix} = \begin{bmatrix} \frac{1}{N^2} E\{\mathbf{1}^T \mathbf{W} (\mathbf{1}^T \mathbf{W})^T\} \\ \frac{4}{N^2} E\{\mathbf{s}^T \mathbf{W} (\mathbf{s}^T \mathbf{W})^T\} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{N^2} \mathbf{1}^T E\{\mathbf{W} \mathbf{W}^T\} \mathbf{1} \\ \frac{4}{N^2} \mathbf{s}^T E\{\mathbf{W} \mathbf{W}^T\} \mathbf{s} \end{bmatrix} = \begin{bmatrix} \frac{1}{N^2} \mathbf{1}^T (\sigma_w^2 \mathbf{I}) \mathbf{1} \\ \frac{4}{N^2} \mathbf{s}^T (\sigma_w^2 \mathbf{I}) \mathbf{s} \end{bmatrix} = \begin{bmatrix} \frac{\sigma_w^2}{N} \\ \frac{2\sigma_w^2}{N} \end{bmatrix} \end{aligned}$$

so the estimators are **consistent**.



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(3) Deriving the Fisher Information Matrix (FIM), we easily find:

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{N}{\sigma_w^2} & 0 \\ 0 & \frac{N}{2\sigma_w^2} \end{bmatrix} \rightarrow \text{CRB}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\sigma_w^2}{N} & 0 \\ 0 & \frac{2\sigma_w^2}{N} \end{bmatrix}$$

$$\text{CRB}(A) = \frac{\sigma_w^2}{N} = \text{MSE}\{\hat{A}_{ML}\}, \quad \text{CRB}(B) = \frac{2\sigma_w^2}{N} = \text{MSE}\{\hat{B}_{ML}\}$$

Hence, the estimators are **efficient**.

The estimator of parameter A has an MSE that is half that of the estimator of B , so A can be estimated with better accuracy. The reason is that A is the amplitude of a (constant) signal that has twice the energy of the sinusoidal signal of which B is the amplitude.

We require: $\sqrt{E\left\{\left(\frac{B - \hat{B}_{ML}}{B}\right)^2\right\}} < \frac{10}{100}$, that implies:

$$E\left\{\left(\frac{B - \hat{B}_{ML}}{B}\right)^2\right\} = \frac{E\left\{(B - \hat{B}_{ML})^2\right\}}{B^2} = \frac{\text{MSE}\{\hat{B}_{ML}\}}{B^2} = \frac{2\sigma_w^2}{NB^2} < \left(\frac{10}{100}\right)^2 \rightarrow N > \frac{200\sigma_w^2}{B^2} = \frac{2000}{4} = 500,$$

so we need to collect at least 500 samples.